Propeller De-icing Timer – Maintenance Practices

1. Adjustment/Test

A. Functional Test Propeller De-icing Timer

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Disconnect connector plug from timer.

3) Connect a jumper wire from pin B of timer receptacle to terminal B of the connector plug, and from pin G of the timer receptacle to ground.

4) Select PROP DE-ICE switch ON.

5) Using a voltmeter, check DC voltage to ground from pin B of the timer. This should be approximately 24 Volt DC.

6) Check DC volts to ground from timer pins C, D, E and F.

Note Pins C, D, E and F are the points at which the system voltage is impressed in sequence to cycle power to the propeller de-icers.Table 1 — Timer Cycling Action shows the cycling action of the timer.

7) Check pins C, D, E and F until a voltage reading of approximately 24 Volt DC is obtained. Hold the voltmeter probe on the pin until the voltage drops to zero. Move the probe to next pin in sequence and check DC volts at each pin. 24 volts should be measured at each pin.

8) Check to ensure that while one pin is energized, that other three pins have zero voltage on them.

9) Select PROP DE-ICE switch OFF.

10) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

11) Remove jumper wire, connect and secure plug to timer.

Table 1 Timer Cycling Action

TIMING SEQUENCE	APPROXIMATE TIME ON*	AREA OF PROPELLER DE-ICERS HEATED
Pin C	34 sec	Right engine propeller, outboard halves
Pin D	34 sec	Right engine propeller, inboard halves
Pin E	34 sec	Left engine propeller, outboard halves
Pin F	34 sec	Left engine propeller, inboard halves
* On time is approximately 34 seconds when operating on normal electrical system voltage. When operating on battery voltage, as in this test procedure, on time may be longer.		

Note The timer does not reposition itself to start at pin C when the system is switched off, but will begin to cycle at the next position from where it was switched off.

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